

On the calibration of the cohesive strength for cohesive zone models in finite element analyses

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ARTICLE INFO

Keywords:

Finite element analysis
Cohesive zone models
Cohesive strength
Beam theory
Calibration

ABSTRACT

Cohesive zone models are commonly used to numerically model fracture phenomena on the microscale. Typical fracture processes are described experimentally via the Double Cantilever Beam (DCB) test and the End Notched Flexure (ENF) test on the macroscale. Analytical solutions, e.g. the LEFM-based beam theory, are used to evaluate these tests and to validate numerical models. However, some input parameters of cohesive zone models, i.e. cohesive stiffness and cohesive strength, are independent of the analytical solutions. In particular, the cohesive strength has a significant effect on the accuracy of the numerical model compared to the analytical solution and thus needs to be calibrated. This work extends the available literature by fitting the numerical to the analytical solution of the DCB and ENF models through calibration of their cohesive strength by using more suitable and generally applicable criteria. It considers the deformation, the applied load and the crack length of the models. The latter has been neglected so far in the existing literature. Via a parametric study, the derived criteria are minimised to find the best fit to the analytical solution. In addition, the effect of all input parameters and of the numerical element size on the calibrated cohesive strength is investigated. The obtained information can be used to qualify numerical models with coarse meshes to accurately compute the fracture of DCB and ENF models. Further, it provides a comprehensive understanding of the cohesive strength as a modelling parameter.

1. Introduction

Adhesive joints have gained importance over the last decades in various application fields with lightweight construction requirements (e.g. automotive, wind turbine blades, aerospace) [1–2]. Another promising application is the interlaminar integration of electronic sensors into fibre-reinforced polymer laminates for Structural Health Monitoring (SHM). These sensors are intended to detect and assess damage to the structure to improve safety in service. Further, this approach can eliminate routine inspection intervals [3–4]. A requirement is that an integrated sensor does not affect the operating limits of the surrounding structure. Thus, the interaction of the sensor and the structure must be estimated in the design phase using numerical finite element analysis (FEA). These numerical results must be validated using experimental data. Common adhesion tests for such purposes are the Double Cantilever Beam (DCB) and the End Notched Flexure (ENF) test. Their evaluation is based on analytical solutions, e.g. the beam theory which is a commonly used method [5–7].

Different modelling approaches are applicable for the simulation of

adhesively joint substrates. In the context of linear elastic fracture mechanics (LEFM), which is the basis for the beam theory, analytical solutions for the fracture process on the macroscale can be derived. It assumes that the nonlinear process zone in the vicinity of the crack tip is small compared to the specimen and can thus be neglected. Additionally, an initial crack is required for propagation and the residual zone behind the crack tip is inaccurately accounted for because all effects are concentrated to the crack tip [2–8–9–10]. Since the LEFM is not able to consider any microscopic stresses or strains around the crack tip, cohesive zone models (CZM) are advantageous. They accurately compute the crack onset and the subsequent propagation on the microscale in FEA models under consideration of local progressive material degradation to account for nonlinear fracture behaviour. This nonlinearity is limited to the process zone. The surrounding body can be still considered as linear-elastic [2–10–11–12]. Thus, in accordance with e.g. Harper and Hallett [13] as well as Soto et al. [14], it is presumed that the macroscopic behaviour of an investigated FEA fracture model can be still described via LEFM. A combination of both approaches allows to consider microscopic, local crack tip phenomena in FEA models via CZM

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<https://doi.org/10.1016/j.tafmec.2022.103733>

Received 8 June 2022; Received in revised form 6 December 2022; Accepted 15 December 2022

Available online 30 December 2022

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and to validate their effect on the macroscopic, global behaviour via LEFM.

A CZM itself is described via a certain shape of a traction-separation law (TSL). In the case of a bilinear TSL, which is used in the current work, three material parameters are required as an input to the model, namely the cohesive stiffness K , the cohesive strength T and the strain energy release ratio (SERR) G . However, only the latter is included in the analytical solution of the above mentioned test processes. The cohesive stiffness K and the cohesive strength T are excluded. Therefore, associated parameter settings cannot be extracted directly from the same test and have to be estimated alternatively. In particular, the cohesive strength has been of great interest in recent literature because the numerical results are very sensitive to this parameter. Ye and Chen [15] used a micromechanical model to predict the cohesive strength for a mixed mode test. Chen et al. [16] applied an inverse analysis to CZM-based simulations to obtain the cohesive zone (CZ) parameters. In the work of Allix et al. [5], the effect of increasing cohesive strength on numerical results for DCB and ENF specimens was studied. Harper et al. [13 17] and Soto et al. [14] varied the cohesive strength in DCB and ENF simulations and compared the obtained results with analytical solutions. In the work of Lu et al. [18], the effect of the cohesive strength on the maximum load was numerically investigated for DCB and ENF models. Whereas an insensitivity was found for fine meshes (0.1 mm element size), a higher sensitivity was concluded for coarser meshes (1 mm element size). Heidari-Rarani et al. [19] presented numerous equations from the literature to obtain the cohesive strength from DCB experiments but outlined the ambiguity about their suitability. Further, by comparing numerical results with experiments, they concluded a significant effect of the cohesive strength on the crack initiation point but not on the crack propagation [19 20]. However, all of these investigations were limited to force–displacement data of the DCB and ENF models. The macroscopic crack extension in terms of the crack length was neglected and no general criterion for evaluation was introduced. Thus, it is currently not possible to obtain the cohesive strength for these single mode fracture models using a defined method that results in high agreement between the numerical model and the analytical solution for both, force and crack length data.

This work addresses the need to calibrate the cohesive strength for DCB and ENF models and overcomes the abovementioned limitations. First, general criteria are introduced to compare the numerical solution with the LEFM-based analytical solution for the DCB and ENF tests [5 13 21 22]. This includes force–displacement data and extends existing approaches by the correlation with progressive damage, i.e. the crack length in particular. Based on these criteria, the presented procedure is intended to fit an arbitrary DCB and ENF model to the beam theory through the calibration of the cohesive strength. The advantage of this procedure is that only macroscopic data are required as a reference to calibrate the cohesive strength as a microscopic parameter. Its significance arises from the fact that the accuracy of the calibration directly defines the accuracy of the numerical solution. In this context, a parametric study for the cohesive strength is performed and yields a calibrated setting for a certain configuration of the model, i.e. for a specific element size as well as individual material and geometry parameters. Therefore, this calibration has to be performed for every model with changed element size, material properties or geometrical design. For this purpose, these parameters are varied in discrete steps to identify their effect on the calibrated cohesive strength. In order to save computational time, this procedure will allow the qualification of coarse mesh FEA models to accurately predict the applied force and the crack length over the duration of the experiment. Only single mode fracture models, i.e. DCB and ENF, are considered in this work. However, an extension to more complex fracture problems, e.g. mixed-mode fracture, is possible insofar as a reference solution exists.

2. Theoretical framework

In this section, basic information about CZM and the analytical solutions for the DCB and ENF tests are provided to enhance the understanding of this work.

2.1. Cohesive zone models

CZM have been widely and successfully used in FEA in recent years for prediction of adhesive joint strength under continuous and fatigue loads [2 23 24]. They were also employed for mixed-mode fracture processes [25 26 27]. Further, numerical models based on CZM were applied to compute different types of materials, such as composite laminates [28 29 30] and reinforced concrete [31 32 33]. All of these studies proved the versatile applicability of CZM for the simulation of various kinds of fracture processes and obtained good results in agreement with e.g. experimental data.

As mentioned earlier, the application of LEFM requires that all nonlinear crack related phenomena are lumped to the crack tip of negligible size. In general, however, a FEA model should consider this nonlinear behaviour on the microscale. Therefore, a nonlinear theory for cracks in terms of CZM must be applied. Equilibrium is achieved by defining crack surfaces with an inherent stress distribution in the crack plane. The definition of the crack surfaces, in particular the computation of the crack front, is the problem to be solved, see Rabczuk [12] and Barenblatt [34]. The nonlinearity is limited to the process zone on the microscale, see e.g. Alfano et al. [10].

A CZM is characterised by specific TSL that relate the local stress t to an associated separation u of the fracture surfaces [5 16]. Via a TSL, crack tip phenomena on the microscale are connected to the macroscopic response of the FEA model [12 23 35]. It is thus highly suitable for the proposed approach in this work. Various shapes of TSL are available in the literature for different application fields, e.g. bilinear, trapezoidal or exponential [2 11 35]. Due to its common application to delamination processes in composite materials and the simple implementation to FEA models [13 27 36], a bilinear TSL is used in this work. However, the presented calibration is not limited to this specific TSL shape.

A bilinear TSL, as shown in Fig. 1a, is described by three parameters K_m , T_m and G_m , which depend on the applied fracture mode (indicated by subscript m). In the DCB test, mode I with crack opening normal to the plane is predominant. The in-plane shear mode II applies for the ENF test, see e.g. Brocks [37] or Sharpe [38]. The cohesive stiffness K_m and the cohesive strength $T_m = t_{max,m}$ characterise the crack onset based on a critical stress or a critical crack tip opening displacement u_c criterion. The SERR G_m is the enclosed area of the TSL and defines the crack evolution [35], see Fig. 1a. While T_m and G_m have significant effects on the numerical results, K_m has just a minor effect. Typically, a fixed value is used for FEA, considering that too small values will apply inappropriate stiffness to the model and too large values will lead to numerical problems due to spurious oscillations [17 35 36].

When the local traction, i.e. the stress t , reaches the cohesive strength T_m , the crack locally starts to propagate and the stress behind the crack front decreases according to the applied TSL, see Fig. 1b. The crack tip is defined as the location where the local traction disappears. The distance between the crack tip and the location of maximum cohesive stress along the interface is defined as CZ length l_{CZ} [21 39 40]. The determination of l_{CZ} using the TSL parameters was subject of several publications, e.g. Turon et al. [35], Harper and Hallett [13] or Soto et al. [14]:

$$l_{CZ} = M \cdot \frac{EG_m}{T_m^2} \quad (1)$$

In Eq. (1), E defines the Young's modulus of the bulk material and M is a scaling factor. The definition of M is of great ambiguity as a wide range can be found in literature based on TSL shape and fracture mode [41 42 43]. Thus, the CZ length in this work must be determined directly

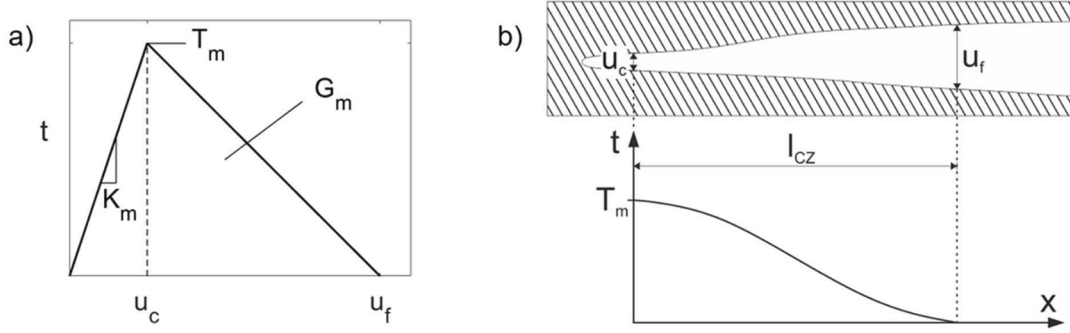


Fig. 1. Schematic visualisation of the cohesive zone via a) the bilinear traction-separation-law and b) the stress distribution for an exemplary mode I crack.

from the cohesive stresses along the interface, see section 3.1.

2.2. Beam theory

Analytical solutions for the DCB and ENF are required for the evaluation of corresponding experimental tests. Therefore, the calibration criteria in this work have to be derived from these equations. The latter are based on the configuration of the DCB and ENF specimens, as displayed in Fig. 2, where L is a characteristic length, a_0 is the length of the precrack, h is the thickness and w is the out-of-plane width of the substrate.

The analytical solutions for the tests are derived from the LEFM-based beam theory. A correction is introduced to account for shear deformation and local deformations around the crack tip, see e.g. Harper and Hallett [13] or Wang and Williams [44]. The correction factors γ and χ , presented by Reeder et al. [45] and Williams [46], are valid for both, DCB and ENF specimens. The equations are adapted to isotropic substrate materials, where E is the Young's modulus and S is the shear modulus:

$$\gamma = 1.18 \cdot \frac{E}{S}, \quad (2)$$

$$\chi = \sqrt{\frac{E}{11S} \left[3 - 2 \left(\frac{\gamma}{1 + \gamma} \right)^2 \right]}. \quad (3)$$

2.2.1. Double Cantilever beam (DCB)

For the analytical description of the DCB test, two sections have to be considered. In the loading section from (O) to (A), as depicted in Fig. 3, the crack length is constant:

$$a = a_0. \quad (4)$$

When the critical displacement d_c is reached, the crack starts to propagate from point (A) to (B) as follows [5 13 22]:

$$a = \sqrt[4]{\frac{9}{4} \frac{d^2}{w} \frac{EI}{G_I}} - \chi h. \quad (5)$$

In Eq. (5), the moment of inertia for rectangular cross sections is defined as $I = wh^3/12$. The critical displacement d_c is obtained from Eq. (4) and Eq. (5). Based on the specific crack lengths for both sections, obtained from the previous equations, the applied force results from [5 13 22]:

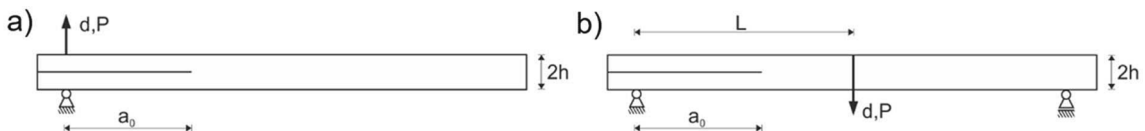


Fig. 2. Configurations of the applied fracture processes: a) DCB and b) ENF (w is the out-of-plane specimen width and not displayed).

$$P = \frac{3}{2} \frac{EI}{(a + \chi h)^3} \cdot d. \quad (6)$$

In accordance with Sharpe [38] and DIN EN 6033 [47], the grey shaded area ΔU under the d - P curve in Fig. 3 is the energy required to propagate a crack of width w over the length Δa . It is defined as the SERR for a mode I loading:

$$G_I = \frac{\Delta U}{w \cdot \Delta a}. \quad (7)$$

2.2.2. End-Notched Flexure (ENF)

The analytical solution of the ENF test consists of three sections, see Fig. 4.

First, from (O) to (A) the crack length retains its initial value a_0 . The critical value of the force P_c results from the definition of the mode II SERR [13] which includes the grey shaded area under the d - P curve:

$$G_{II} = \frac{9(a_0 + 0.42\chi h)^2}{2L^3 + 3(a_0 + 0.42\chi h)^3} \cdot \frac{1}{w} \cdot \frac{1}{2} P_c d_c. \quad (8)$$

The applied force linearly increases with the centre displacement d [5 13 21]:

$$P = \frac{8Ewh^3}{2L^3 + 3(a_0 + 0.42\chi h)^3} \cdot d. \quad (9)$$

When the critical load P_c is reached, the crack length jumps from (A) to (B), which correlates with an abrupt drop of the applied force to a lower level P^* . The corresponding crack length a^* at $d = d_c$ is [13 21]:

$$a^* = \sqrt[3]{\frac{8}{3} \frac{Ewh^3}{P^*} - 2L^3}. \quad (10)$$

In the last section, from (B) to (C) and for $a \geq L$, the change of crack length Δa is inversely proportional to the SERR, where ΔC is the change of the specimen compliance $C = d/P$ [8 9 48]:

$$G_{II} = \frac{\Delta C}{\Delta a} \frac{P^2}{2w}. \quad (11)$$

Furthermore, the displacement in the last section is related to the applied force as follows [5 21]:

$$d = \frac{P}{24EI} \left[2L^3 - \frac{(64G_{II}wEI)^{\frac{3}{2}}}{4\sqrt{3}P^3} \right]. \quad (12)$$

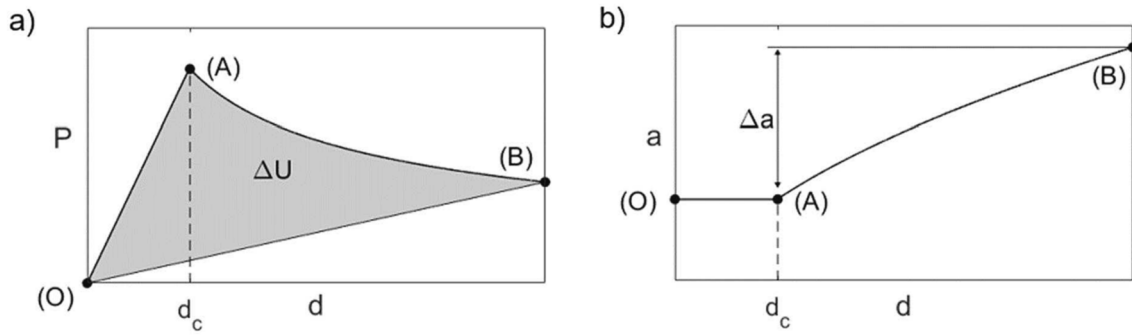


Fig. 3. Analytical solution for the DCB test: a) force-displacement data, b) crack length-displacement data.

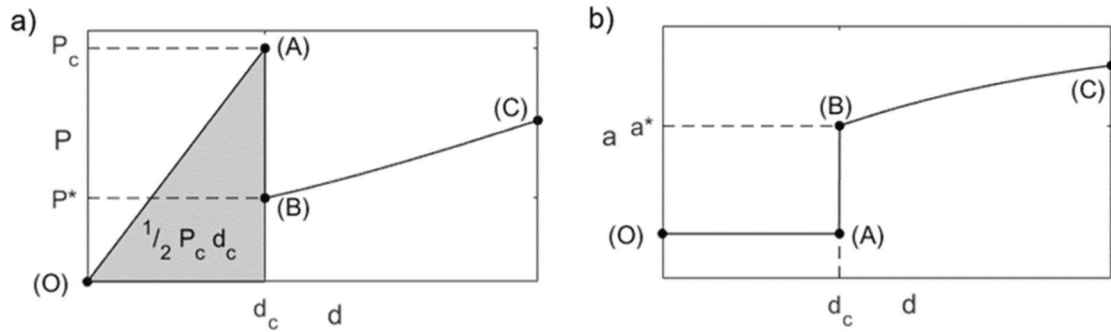


Fig. 4. Analytical solution for the ENF test: a) force-displacement data, b) crack length-displacement data.

The criterion for stable crack growth $\Delta G/\Delta a \leq 0$ is only met in the ENF configuration for crack lengths $a \geq 0.7L$ [5 48 49]. This crack length is exceeded during the force drop. As a result, the cohesive zone can develop. This issue is discussed in the next section.

It is emphasized again that G_m is the only quantity of the TSL which finds its way into the above equations. The other quantities T_m and K_m do not occur.

3. Numerical setup

In this section, information about the setup of the numerical models are given. Furthermore, general criteria for evaluating the fit between the numerical solution and the beam theory are introduced. Finally, all parameter settings for the parametric study are listed.

3.1. Finite element models

The numerical 2D models are generated using the FEA software ABAQUS. Both models, in DCB and ENF configuration, are basically the same and differ only with respect to their specific boundary conditions, see Fig. 2.

In a first step, two substrates are modelled as rectangular 2D solids with a spacing of 0.001 mm. Isotropic material properties with a Young's modulus E , that is varied during the parametric study, and a fixed Poisson's ratio of 0.3 are assigned to the substrates. Meshing is performed using eight-node plane stress elements with reduced integration (CPS8R). Hence, the distance between two adjacent integration points l_P is half of the element size.

A cohesive interaction is modelled to apply the bonded interface between the substrates. A surface-to-surface discretisation with a small sliding formulation is selected. The bottom edge of the upper substrate is considered the master surface, the top edge of the lower substrate is considered the slave surface, see Fig. 5. Bonding is limited to an editable set of nodes on the slave surface. Using this method, different precrack lengths a_0 can be applied to the model. In the next step, the CZ properties K_m , T_m and G_m as well as the specimen width w are defined for the interaction. For this work, the cohesive stiffness has a value of $K_m = 1e15 \text{ N/m}^3$. The magnitude of K_m is retained in all simulations since its effect on the solution is negligible [17 36]. Damage initiation is defined by a maximum nominal stress criterion. In accordance with the applied TSL, an energy-based damage evolution with linear softening is selected. Since just pure mode I and mode II fracture is investigated, no mixed

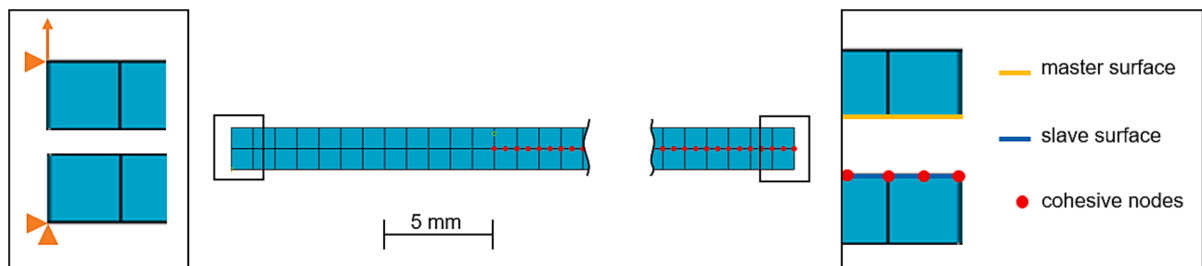


Fig. 5. Schematic representation of the numerical model in DCB configuration indicating the boundary conditions on the left side and the application of the cohesive interaction on the right side.

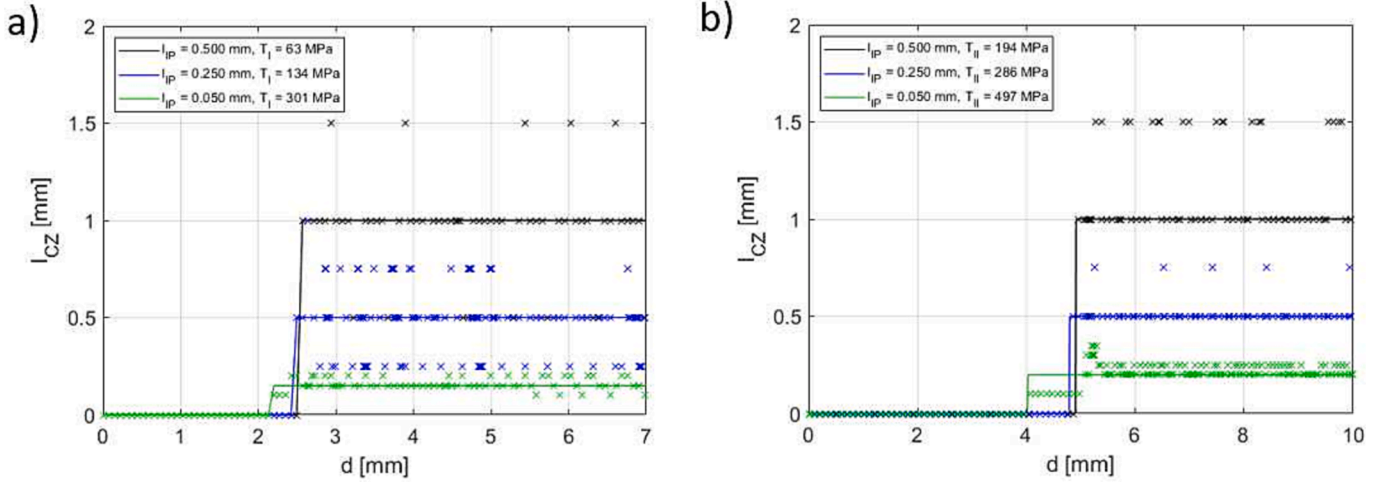


Fig. 6. Obtained cohesive zone length l_{CZ} for exemplary l_{IP} and T_m in dependency of the applied displacement in the a) DCB and b) ENF configuration.

mode parameters are required. No additional contact interactions, i.e. normal and tangential behaviour, are considered since no effect on the numerical solution was obtained in preliminary studies.

Individual boundary conditions and continuously increasing displacements are applied to the models (see Fig. 2). The maximum displacement for the DCB configuration is 15 mm and 10 mm for the ENF case. The displacement is applied in increments of maximum 0.01 which is automatically reduced during crack propagation with an allowed minimum of 10^{-45} . No non-linear geometry option is employed to improve comparability with the LEFM-based analytical solution.

Finally, the following data sets are required for the model output: the reaction force at the same node where the displacement is applied, as well as the bond states and the local cohesive stresses t_m for the node set on the slave surface. The bond state has the value 0 ($t_m = 0$) or 1 ($t_m \neq 0$). For evaluation of the CZ length, the crack tip position is determined from the bond state data ($t_m(x) = 0$ in Fig. 1b) and the location of maximum stress ($t_m(x) = T_m$) is determined from the cohesive stress data. The resulting value for l_{CZ} is defined as the mean value of all recorded CZ lengths from the crack onset until the end of the simulation, see Fig. 6. It can be seen that the CZ forms after the crack starts to propagate. In the following, the CZ length varies in the magnitude of the integration point distance l_{IP} , but the formation of a plateau is clearly visible. Thus, a steady state of the crack propagation with a constant toughness can be concluded that allows reliable determination of the SERR [19 50].

3.2. Evaluation criteria

Since K_m is assumed to be fixed, T_m is the only unknown parameter of the numerical model so far that is not included in the beam theory. Thus, T_m has to be calibrated. A numerical simulation with a calibrated cohesive strength $T_{m,cal}$ should reproduce the beam theory. The nominal value $G_{m,nom}$ of the SERR (see Table 1) is used as an input for the

simulation and the beam theory, whereas $G_{m,eff}$ is the effective SERR that is calculated from the output of the simulation via Eq. (7), respectively Eq. (7). For a high level of agreement between numerical model and beam theory, the relative error.

$$f_G = \frac{G_{m,eff} - G_{m,nom}}{G_{m,nom}} \quad (13)$$

has thus to be minimised in absolute terms.

Since f_G can have multiple local extreme points, two further criteria and boundary conditions are introduced. (I) The CZ length resulting from the calibrated cohesive strength must be larger than the distance between two adjacent integration points l_{IP} to avoid potential errors due to the model discretisation. (II) As T_m affects the critical displacement for crack onset d_c [5 51], the calibrated value has to match the theoretical crack onset properly.

3.3. Parameter settings

The reference configurations for DCB and ENF specimens are highlighted in Table 1. The settings for the material parameters E and $G_{m,nom}$ as well as for the geometry parameters are chosen to match experimental measurements, e.g. as conducted in the work of Genco et al. [52]. Thus, different parameter settings apply for the DCB and the ENF configuration. A coarse mesh with an integration point distance l_{IP} of 0.5 mm is used as a reference to save computation time. The mesh is stepwise refined to investigate the effect of the integration point distance on the resulting error f_G . To study the effect of all numerical input parameters included in the beam theory on the calibrated cohesive strength $T_{m,cal}$, the parameters are varied within a reasonable range. While the effect of one parameter is investigated, including the integration point distance, all other parameters are kept constant in reference setting.

For determination of $T_{m,cal}$, the numerical simulations are run several

Table 1

Settings for numerical input parameters (NIP), reference values are highlighted in bold. The DCB solution is independent of L , therefore no parametric study is conducted for this case.

NIP	unit	DCB	ENF
l_{IP}	[mm]	[0.050, 0.125, 0.250, 0.375, 0.500]	[0.050, 0.125, 0.250, 0.375, 0.500]
E	[GPa]	[22.5, 25.0, 27.5, 30.0, 32.5]	[40.0, 42.5, 45.0 , 47.5, 50.0]
$G_{m,nom}$	[N/m]	[650, 800, 950 , 1100, 1250]	[1750, 2000, 2250 , 2500, 2750]
a_0	[mm]	[8, 10, 12, 14, 16]	[6, 8 , 10, 12, 14]
L	[mm]	[-]	[21, 23, 25 , 27, 29]
w	[mm]	[11, 13, 15 , 17, 19]	[11, 13, 15 , 17, 19]
h	[mm]	[0.80, 0.85, 0.90, 0.95 , 1.00]	[0.80, 0.85, 0.90, 0.95 , 1.00]

times with increasing cohesive strength. For the DCB configuration, T_I is varied from 1 MPa to 300 MPa in steps of 1 MPa. For the ENF configuration, the step size of T_{II} is 2 MPa as the interval is increased from 100 MPa to 700 MPa.

4. Numerical results

In this section, the numerical results obtained are presented and discussed. The calibrated cohesive strength $T_{m,cal}$ is derived for all configurations and parameter settings using the introduced criteria. After each simulation has been completed, Eq. (7) is applied for the evaluation in DCB configuration, respectively Eq. (8) for ENF configuration. Finally, the relative error f_G between the nominal SERR $G_{m,nom}$ and the effective value $G_{m,eff}$ is obtained via Eq. (13).

4.1. Effect of cohesive strength on the relative error f_G

In the first step, the focus is on the reference configurations for the DCB and ENF tests. The effect of an increasing cohesive strength on the relative error f_G is evaluated for different integration point distances l_{IP} , see Fig. 7. For ease of comparison, the scale of the ordinate axis (f_G) is kept constant. Data are only considered when the resulting value of l_{CZ} is larger than l_{IP} . For that reason, no local minimum above $T_I = 63$ MPa can be found in Fig. 7 for the case that $l_{IP} = 0.500$ mm. Thus, according to Eq. (1), the cohesive strength must not approach infinity since l_{CZ} would tend to zero in that case. On the contrary, a setting for $T_{m,cal}$ [$0 < T_{m,cal} < \infty$] needs to be determined that meets the introduced criteria for a certain set of applied parameters.

It is clearly evident that f_G is always positive for DCB configuration, except for a few points for very small T_I . Therefore, the effective value $G_{I,eff}$ of the numerical model tends to overestimate the nominal SERR $G_{I,nom}$ since the reduction of ΔU due to inappropriate settings of T_I is found larger than the reduction of Δa , see Eq. (7) and Fig. 8. For the ENF test, the relative error is negative for small T_{II} and becomes positive for large cohesive strengths. The reason for this general trend is evident from Eq. (8). Increasing T_{II} cause larger P_c (see Fig. 9), which is in accordance with e.g. Lu et al. [18], and thus increase the effective SERR $G_{II,eff}$.

Another important finding is that the function f_G over T_m has no continuous slope. Multiple local minima and maxima can be identified for both configurations. This is due to the dependence of the CZ length and the cohesive strength according to Eq. (1). A correlation of the error f_G with the associated mean CZ length l_{CZ} of each simulation reveals that negative slopes in Fig. 7 refer to a smooth transition of l_{CZ} from a certain multiple of l_{IP} to the next lower. Conversely, l_{CZ} is almost constant where f_G has a positive slope. Thus, local maxima form directly before such a transition and local minima after the transition. These conclusions are confirmed by the different amplitudes and positions of the local extreme

points for various element sizes. The distance between two integration points decreases for finer meshes. Thus, the amplitudes of f_G become smaller because the distance between adjacent multiples of l_{IP} decreases in parallel and the transition is less pronounced.

This is well observed for $l_{IP} = 0.050$ mm. In both configurations, DCB and ENF, local extreme points are hard to identify for small T_m because large CZ lengths can be modelled well. As the cohesive strength increases, l_{CZ} decreases and approaches l_{IP} . Local extreme points are formed because certain l_{CZ} can be potentially not realised properly by the model.

Thus, two correlations that are not fully independent from each other affect the error f_G as a function of T_m for the DCB and ENF configuration: the effect of the cohesive strength on the ultimate load and on the CZ length. A calibrated setting $T_{m,cal}$ must therefore consider both effects and the error f_G should be close to zero. For the DCB configuration this is the case for local minima. In ENF configuration, the absolute value of f_G is small at local extreme points and for the zero crossing.

A difference between the DCB and the ENF configuration is the magnitude of f_G for several local extreme points. In the DCB test, the relative errors of the minima are constant in the range of 2.14% or lower ($l_{IP} = 0.500$ mm). In contrast, the associated errors of the local maxima and the zero crossing for the same integration point distance range from -17.33% to -0.02% in the ENF configuration. For a better understanding of the effect of T_m on the numerical solution, these characteristic values of the cohesive strength for $l_{IP} = 0.500$ mm are plotted in Fig. 8 (DCB) and Fig. 9 (ENF). The largest investigated integration point distance is selected for the visualisation since the primary intention of this procedure is to qualify coarse mesh models for application.

For the DCB configuration, the relative errors of the minima are found constantly positive and small. However, a large deviation from the beam theory is seen for small $T_{m,cal}$. The released energy ΔU (see Fig. 3a) is computed much too small. Concurrently, the crack onset displacement d_c increases for small cohesive strengths, resulting in a small change in crack length Δa . Both effects cancel each other out. Therefore, according to Eq. (7), the SERR of the simulation has a low error compared to the nominal value. To ensure a good fit with the beam theory, in terms of a small relative error f_G and a good agreement with d - P and d - a data, the largest possible $T_{I,cal} = 63$ MPa must be used which satisfies the requirement of $l_{CZ} \geq l_{IP}$ and computes d_c with highest possible accuracy. Since higher values for $T_{m,cal}$ reduce the l_{CZ} in accordance with Eq. (1), the prerequisite for the application of LEFM and thus the beam theory is met with higher probability. However, a further increase of T_I above $T_{I,cal} = 63$ MPa would favour the overestimation of both, the ultimate load P_c and the crack initiation displacement d_c , see Lu et al. [18].

Focusing only on the d - P data, as done in the recent literature, confirms that the data for all T_I converge the beam theory during crack propagation ($d > d_c$). However, the d - a data deviate significantly from

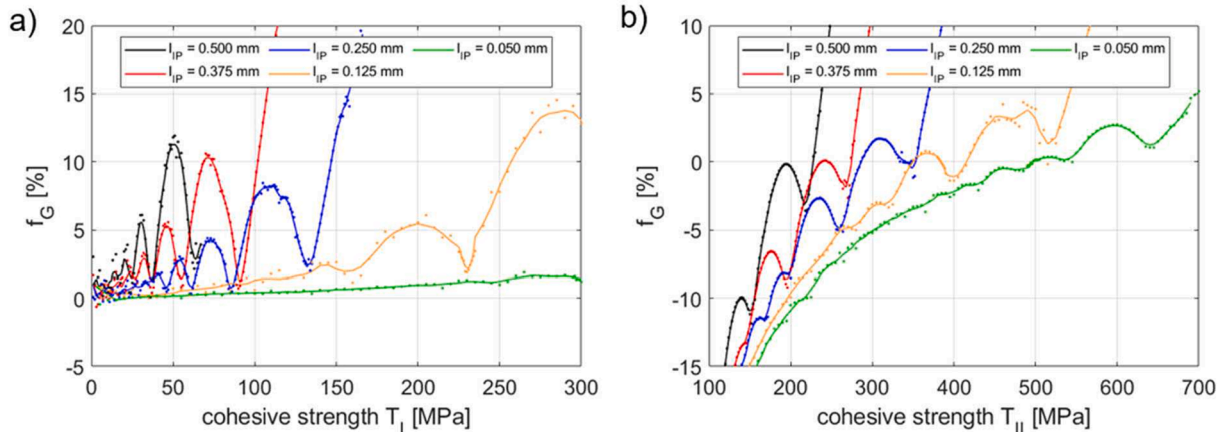


Fig. 7. Resulting relative errors f_G for increasing T_m and various l_{IP} in a) DCB configuration and b) ENF configuration.

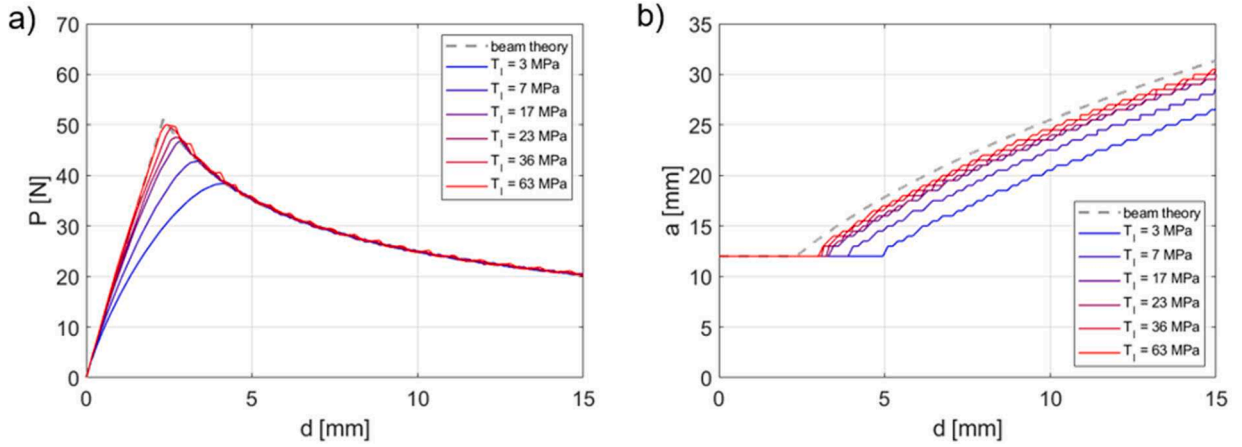


Fig. 8. Effect of local minima T_I on the results for the DCB configuration with $l_{IP} = 0.500$ mm: a) force-displacement data, b) crack length-displacement data.

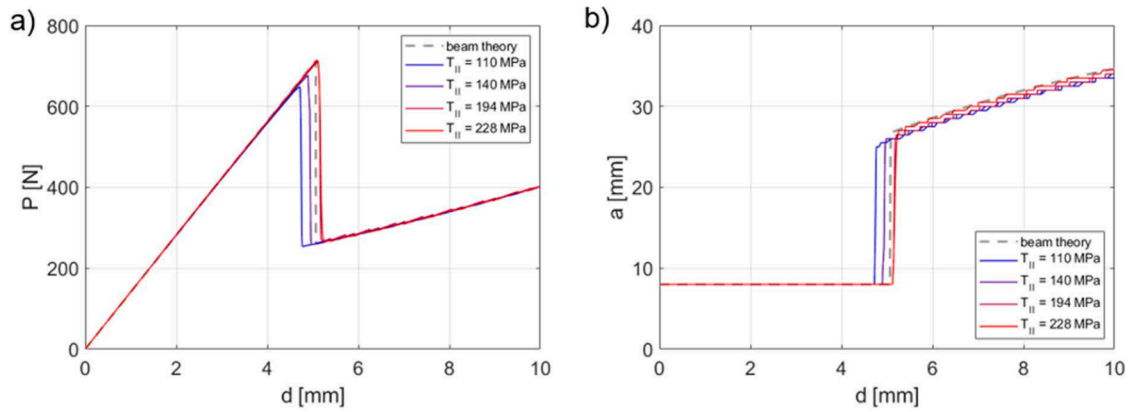


Fig. 9. Effect of local maxima T_{II} and the zero crossing on the results for the ENF configuration with $l_{IP} = 0.500$ mm: a) force-displacement data, b) crack length-displacement data.

the beam theory. This proves that only a comprehensive evaluation of f_G and d_c can lead to a calibrated cohesive strength $T_{I,cal}$ for the DCB test.

In contrast, small values for T_{II} in the ENF configuration (Fig. 9) underestimate d_c and too large values overestimate the crack onset. Since the initial compliance before crack onset ($d < d_c$) is almost unaffected by the cohesive strength, the crack onset displacement is a direct measure for f_G because G_{II} depends linearly on it, see Eq. (8). This results in negative errors for small T_{II} and positive errors for large values. Thus, the smallest error of f_G ensures optimal prediction of d_c . For the displayed l_{IP} of 0.500 mm, $T_{II,cal}$ is equal to 194 MPa. Other integration point distances result in varying values for $T_{II,cal}$.

4.2. Effect of input parameters on $T_{m,cal}$

As concluded, the element size has an effect on the calibrated cohesive strength. The findings are important for the qualification of coarse mesh FEA models. Moreover, it is of great interest for future models to investigate the effect of all numerical input parameters on $T_{m,cal}$. Computational effort for the calibration can be saved with a priori knowledge about their correlation.

In Fig. 10, the calibrated cohesive strength is plotted against the numerical input parameter settings (see Table 1). The data for the DCB test are displayed in black, the ENF data are shown in blue. The limits of the ordinate axis ($T_{m,cal}$) are equal for all subfigures for ease of comparison. The mean relative error f_G is 1.56% for $T_{I,cal}$ and 0.21% for $T_{II,cal}$. The higher error for the DCB configuration results from the additional consideration of a matching d_c . Thus, the calibrated cohesive

strength does not necessarily coincide with the lowest f_G .

Three groups of parameters can be identified from Fig. 10: parameters with no effects (L , w), parameters with small effects (a_0 , h) and parameters with large effects (l_{IP} , E , $G_{m,nom}$).

A comparison of the calibrated cohesive strength values between DCB and ENF configurations is not meaningful. For the DCB configuration, $T_{I,cal}$ represents a critical normal stress and $T_{II,cal}$ is a critical shear stress for the ENF test. However, cohesive strength values for the latter are generally higher, favoured by higher settings of E and $G_{II,nom}$ according to Eq. (1).

Generally, the trends are the same for both tests. An increase in the stiffness of the specimen (i.e. large E , large $G_{m,nom}$, large h and small a_0) requires an increase of the cohesive strength $T_{m,cal}$. For the material parameters E and $G_{m,nom}$, this correlation is evident from Eq. (1), assuming a constant CZ length l_{CZ} which is always a multiple of l_{IP} . The effect of the thickness h has a similar cause. A large substrate thickness increases the bending stiffness EI with $I = wh^3/12$, see sections 2.2.1 and 2.2.2. It has thus an equivalent effect on the numerical simulation as an increased Young's modulus E . This correlation cannot be seen for the width w since it is just a scaling parameter in the 2D model. However, an increasing effect of w on $T_{m,cal}$ is expected for 3D models.

The largest observable effect results from the integration point distance l_{IP} . Using coarse meshes with large l_{IP} limits the accuracy of the numerical model to large CZ lengths resulting from small cohesive strengths. Reducing the element size favours the simulation of smaller CZ lengths caused by higher T_m .

The initial crack length a_0 is the only parameter with slightly

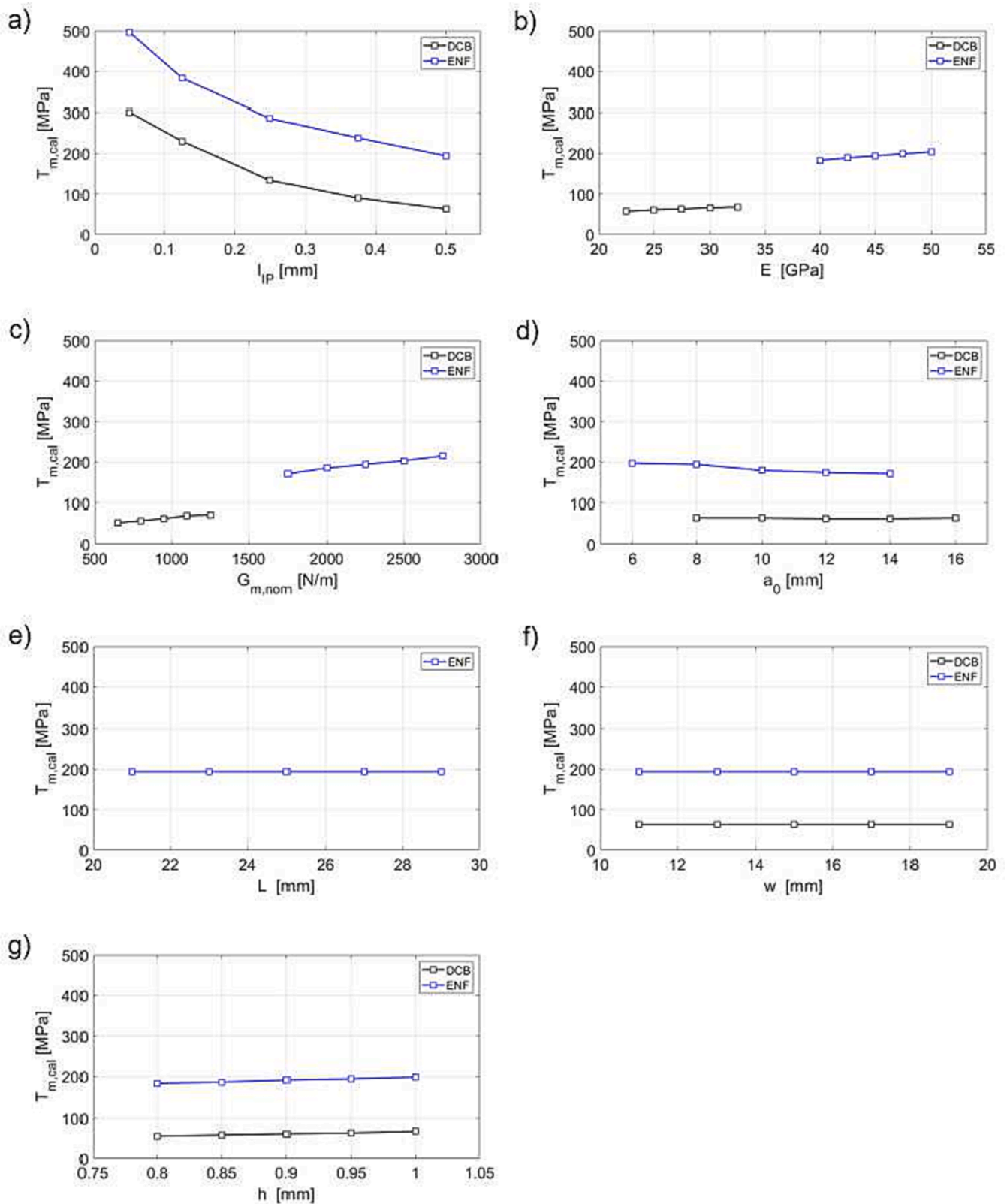


Fig. 10. Effects of numerical input parameter settings on $T_{m,cal}$ for DCB (black) and ENF (blue) configuration.

different effects. For the DCB configuration, $T_{I,cal}$ varies between 63 MPa and 65 MPa for the settings studied. The effect is found to be larger for the ENF test. The cohesive strength $T_{II,cal}$ ranges from 171 MPa to 197 MPa, but shows a noticeable increase for small values of a_0 . A reason for this could be that the accuracy of the numerical model is observed to be

limited to an a_0/L -ratio between 0.2 and 0.6. Extremal settings of a_0 are close to these boundaries.

Any change in the element size and the material parameters requires a new calibration of the applied cohesive strength. Changes to the geometric configuration of the specimen have just small effects on $T_{m,cal}$.

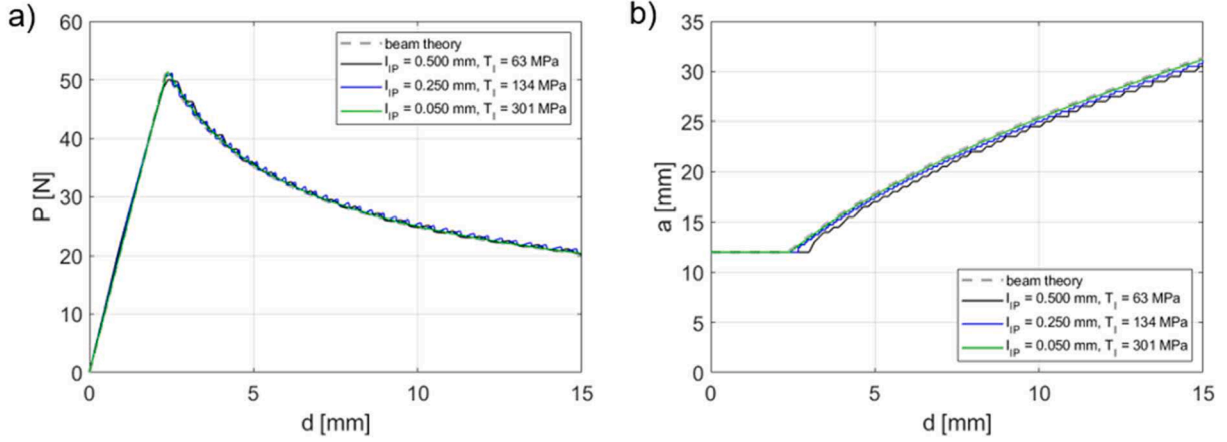


Fig. 11. Numerical results for $T_{I,cal}$ of various integration point distances l_{IP} : a) force-displacement data, b) crack length-displacement data.

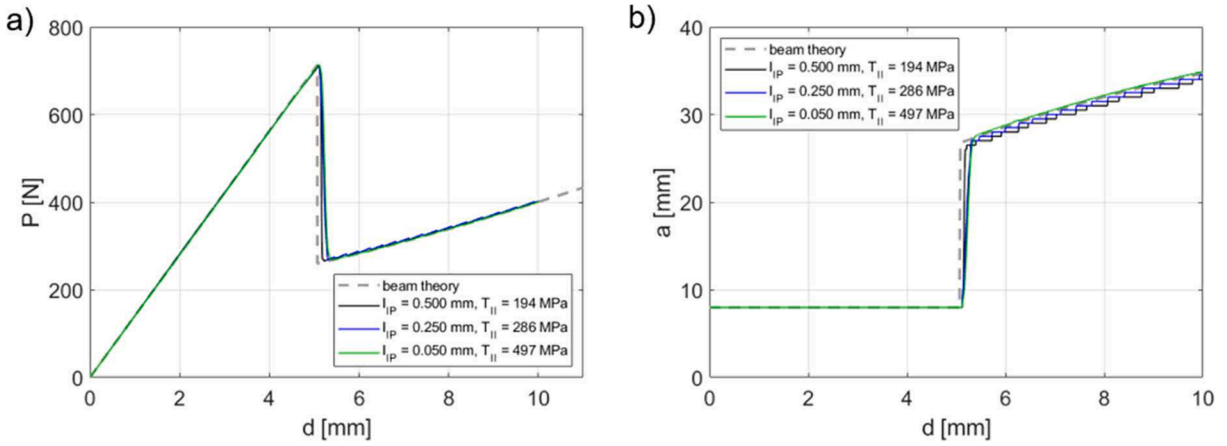


Fig. 12. Numerical results for $T_{II,cal}$ of various integration point distances l_{IP} : a) force-displacement data, b) crack length-displacement data.

However, a calibration is of great importance because even small deviations from the calibrated cohesive strength have large effects on the accuracy. As shown in Fig. 7, the relative error f_G increases rapidly near the optimum. Therefore, an accurate model requires accurate calibration of the cohesive strength, especially for coarse meshes.

4.3. Consistency of $T_{m,cal}$ with the beam theory

Finally, numerical simulations with the determined $T_{m,cal}$ are plotted in Fig. 11 (DCB) and in Fig. 12 (ENF) for the reference parameter settings and directly compared to the beam theory. In addition to the coarse mesh ($l_{IP} = 0.500$ mm), data sets for the medium ($l_{IP} = 0.250$ mm) and the fine mesh ($l_{IP} = 0.050$ mm) are added. The individual $T_{m,cal}$ for every element size is applied.

The analytical solution for the DCB test based on the beam theory can be found to be highly consistent with the numerical results with respect to the d - P data, regardless of the applied element size. With respect to the d - a data, a small effect can be identified. For the coarsest mesh, the crack onset is still slightly overestimated because the cohesive strength T_I cannot be further increased due to limitations of the CZ length. However, this also depends on other parameter settings. The initial offset to the beam theory for the d - a data remains constant until the end of the simulation. This highlights the importance of accurately estimating d_c . Almost no difference can be observed for the refined meshes.

As mentioned above, the relative error f_G directly coincides with the error for d_c in the ENF test. Thus, both errors are minimised in parallel. This can be seen in Fig. 12. Both, d - P data and d - a data fit the beam

theory accurately for all element sizes. Minor deviations relate to the slope of the force drop and the jump of the crack length. Additionally, fine meshes increase the smoothness of the crack propagation because more integration points are available along the CZ. However, coarse meshes still compute the crack propagation with high accuracy.

The presented plots prove that a calibrated cohesive strength, considering small errors for f_G and d_c , can accurately model the analytical solutions for the DCB and ENF test. Therefore, the introduced criteria prove to be applicable and reliable. Also coarse meshed models show large consistency with the theoretical reference.

5. Conclusions

In this work, an approach is presented to calibrate the cohesive strength T_m of CZM to match the beam theory for the DCB and ENF test. The need for calibration was derived from the fact that the equations of the beam theory, that are commonly used for test evaluation, are independent of this parameter. Therefore, in extension to recent literature, a general calibration criterion in terms of the relative error f_G was introduced that is applicable for both tests. Further, the evaluation was always performed considering an accurate estimation of the crack onset displacement d_c . This approach extends previous works where the crack propagation was not a criterion for the fit of the numerical solution.

It was found that the criterion for f_G has several extreme points which are closely related to the accurate prediction of the cohesive zone length. The amplitudes of f_G decrease with the element size. Additionally, the significance was shown to capture the theoretical crack onset

displacement. For the DCB test, it was introduced as an additional calibration criterion. For the ENF test, it was proved to coincide with a small error for f_G .

In the next step, the introduced criteria were used to study the effect of all numerical input parameters on the calibrated cohesive strength $T_{m,cal}$. For this purpose, the parameters were varied in a reasonable range. A large effect was observed for the element size and the material parameters. Geometrical parameters of the specimen had small or even no effect on $T_{m,cal}$. In general, the effects for both tests had similar trends.

The calibrated settings for the cohesive strength have shown a high level of consistency with the beam theory, regardless of the applied element size.

This work aims to extend the understanding of the cohesive strength as a modelling parameter. The introduced criteria and the consideration of d - a data in addition to the d - P data are important improvements for the calibration procedure. It has been shown that even coarse meshed DCB and ENF models are capable of computing the beam theory on the macroscale with relative errors of 2% or lower. These findings will help to save computational effort for further simulations, in 2D and especially 3D, by qualifying coarse meshes. The requirement for qualification can be met by the calibration procedure presented without the need for microscopic data related to the cohesive zone. The future focus will be on the extension of the presented approach to 3D problems of the DCB and ENF configurations. The determination of the cohesive strength for a wide range of material properties, i.e. the Young's modulus and the SERR, allows the definition of a regression function. It can be potentially used to omit the calibration for every discrete parameter combination in order to save time and to achieve a higher degree of applicability. Further, the presented calibration procedure allows for the application to more complex TSL shapes or mixed-mode fracture.

CRedit authorship contribution statement

Max Linke: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Rolf Lammering:** Supervision, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the Deutsche Forschungsgemeinschaft [grant number 3933868053].

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